AMAZON FOOD REVIEW SENTIMENT ANALYSIS

BUSINESS PROBLEM

Many businesses receive thousands of customers reviews every day. Manual review analysis is time-consuming. This project automatically classifies reviews into Positive, Neutral and Negative sentiments and extracts business insights.

PROJECT GOAL

Build an NLP-based sentiment analysis solution for Amazon Food Reviews.

OBJECTIVES

Analyze reviews, clean text, perform EDA, apply NLP, classify sentiment, visualize findings, and generate business recommendations.

DATASET

Amazon Fine Food Reviews (568,454 reviews).

TOOLS

Python, Pandas, NumPy, Matplotlib, Seaborn, NLTK, WordCloud, Scikit-learn.

DATA PREPROCESSING

Missing value check, duplicate check, sentiment mapping, lowercase conversion, HTML removal, URL removal, number removal, punctuation removal, stopword removal, lemmatization.

EDA

Ratings distribution, sentiment distribution, pie charts, review length analysis, boxplots, heatmap.

NLP

Tokenization, stopword removal, lemmatization, VADER sentiment analysis, TF-IDF feature extraction.

MACHINE LEARNING

Multinomial Naive Bayes with TF-IDF features, train-test split, prediction and EVALUATION.

MODEL PERFORMANCE

Accuracy ≈ 81.8%. Positive sentiment dominated the dataset.

BUSINESS INSIGHTS

Positive reviews praise quality and taste. Negative reviews mention packaging, stale products, delayed delivery and freshness.

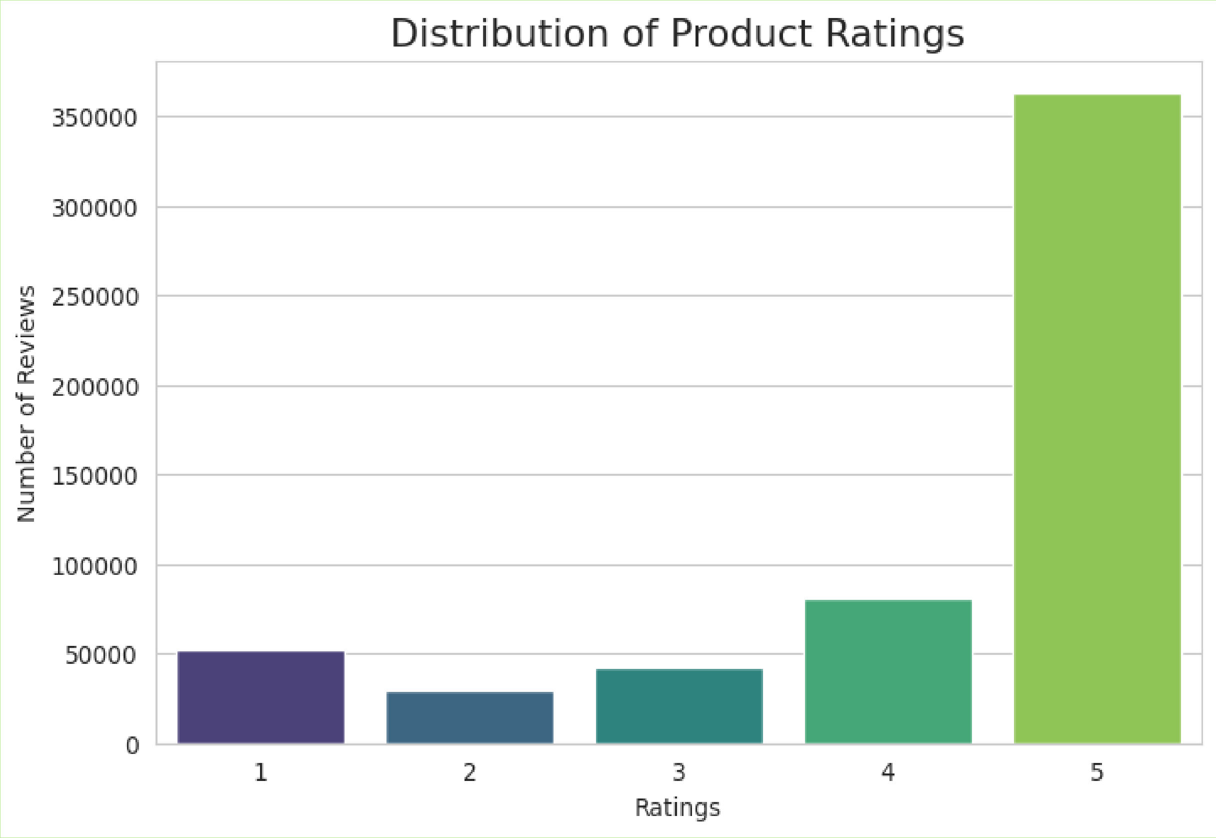
RECOMMENDATIONS

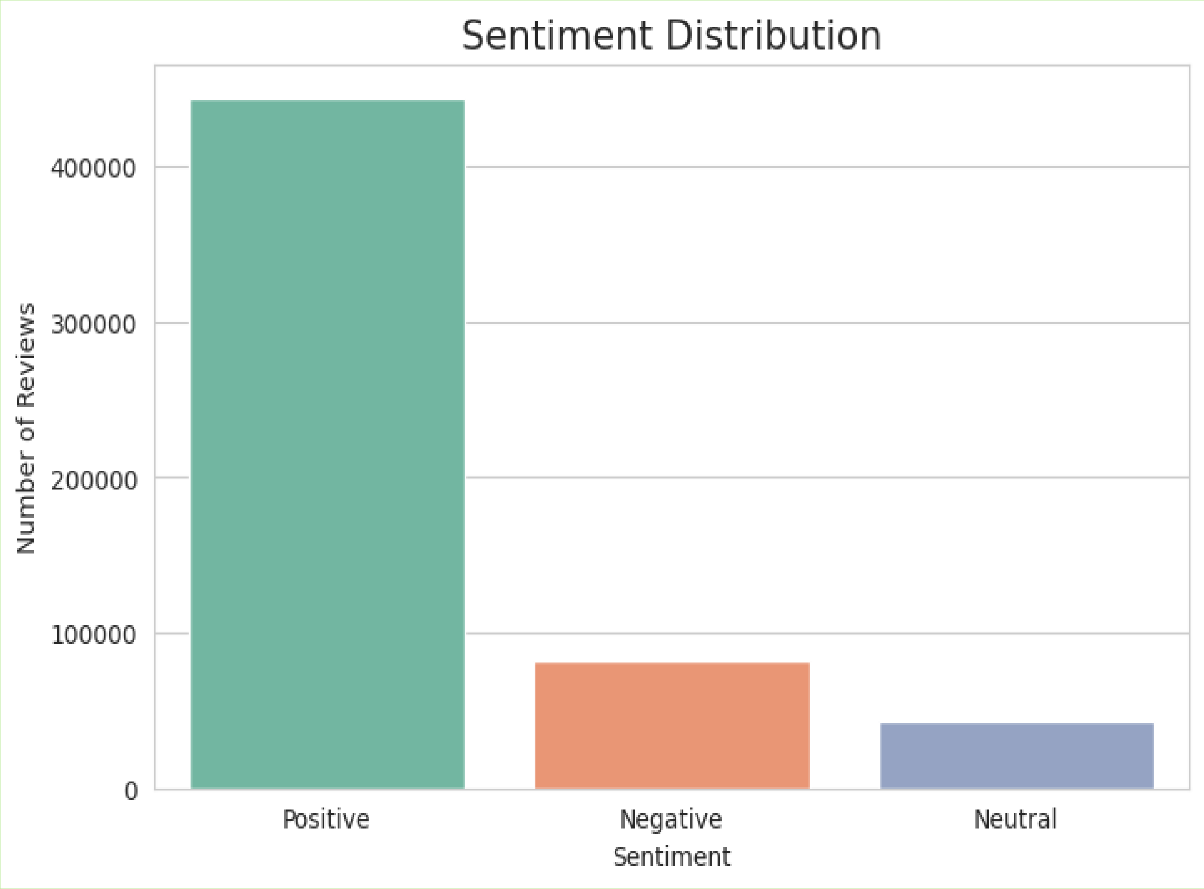
Improve quality control, packaging, logistics, customer service and continuously monitor customer sentiment.

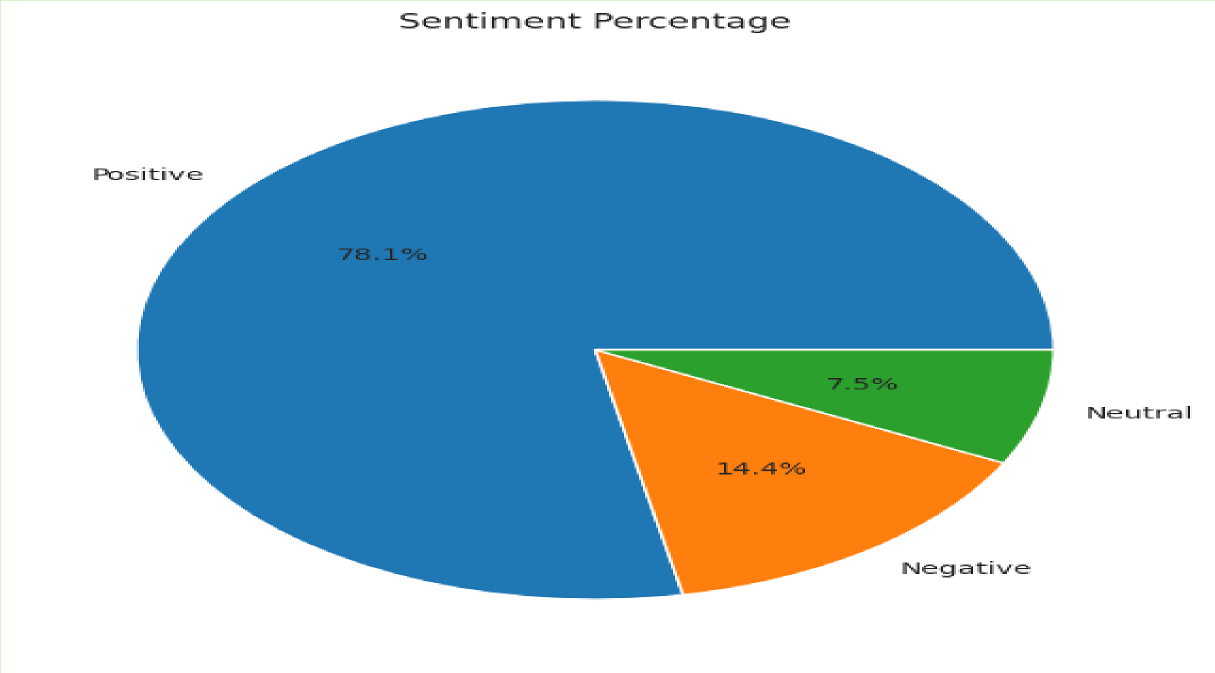
CONCLUSION

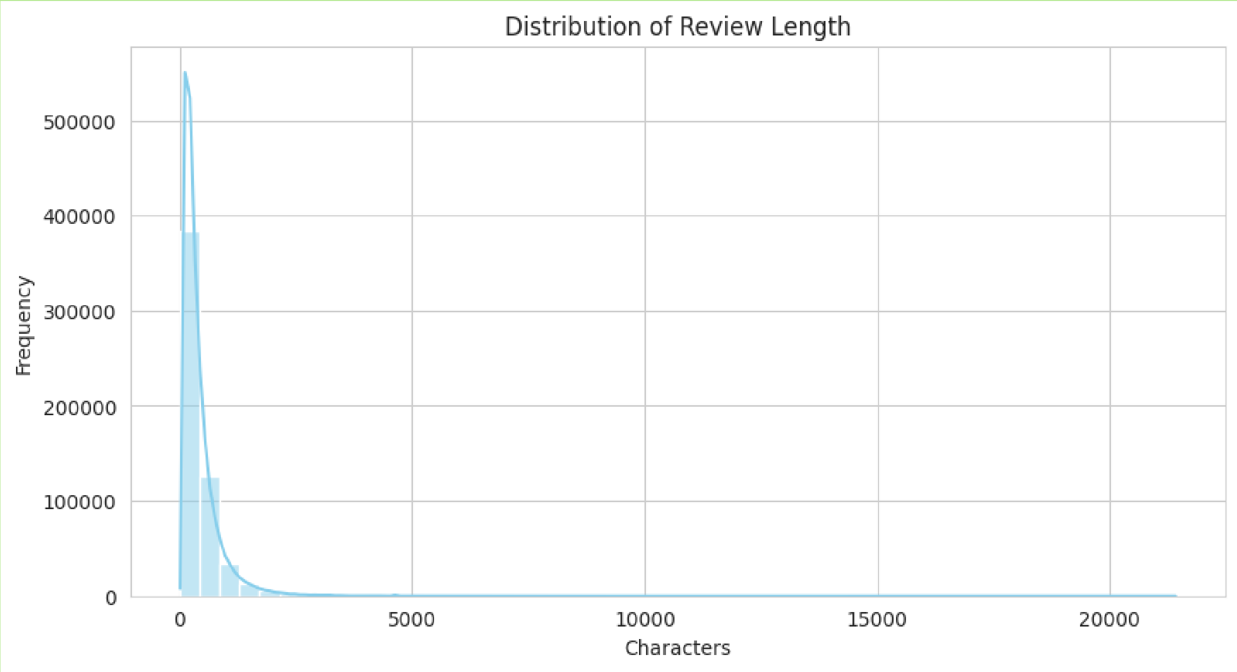
The project demonstrates how NLP converts customer feedback into actionable business intelligence supporting marketing, product improvement and customer experience.

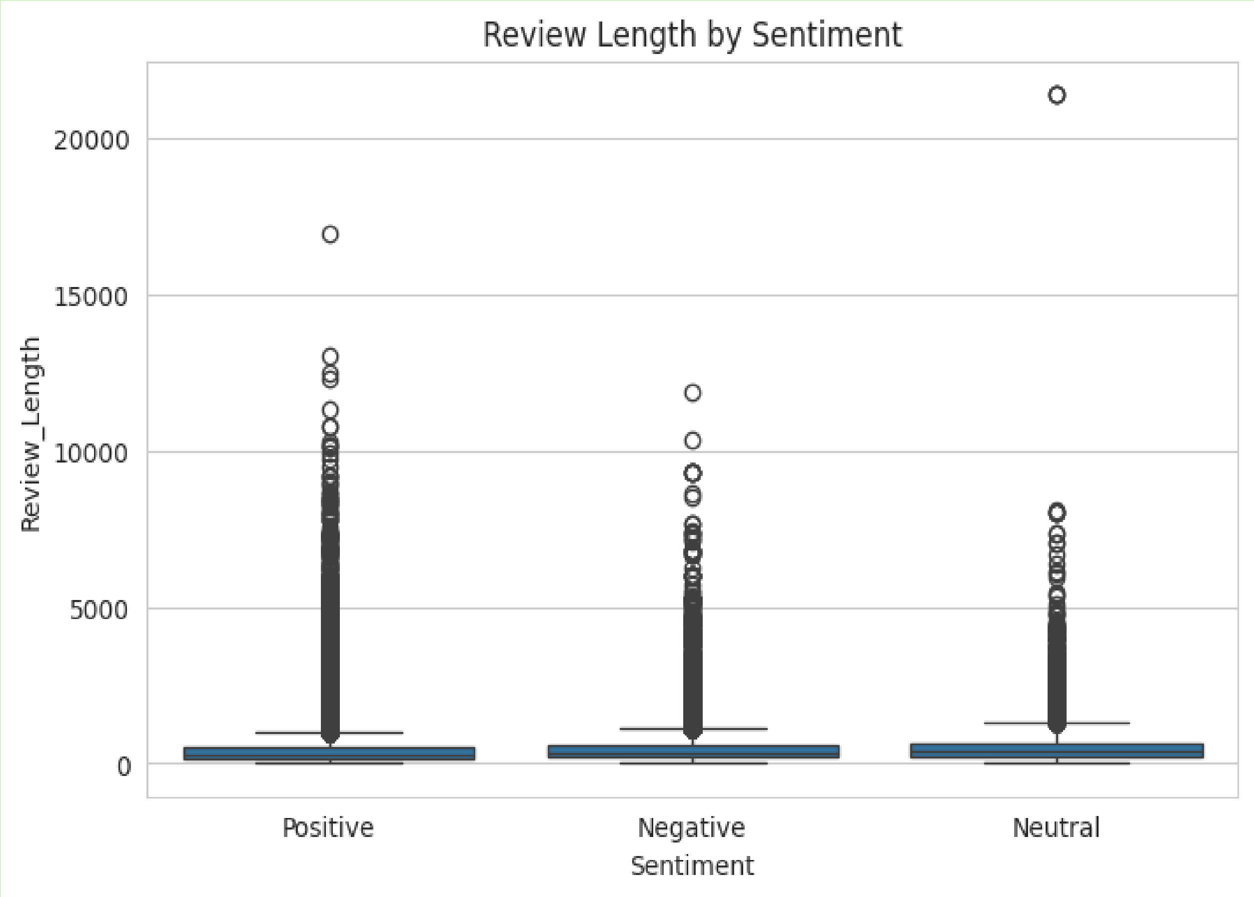
CHARTS AND GRAPHS

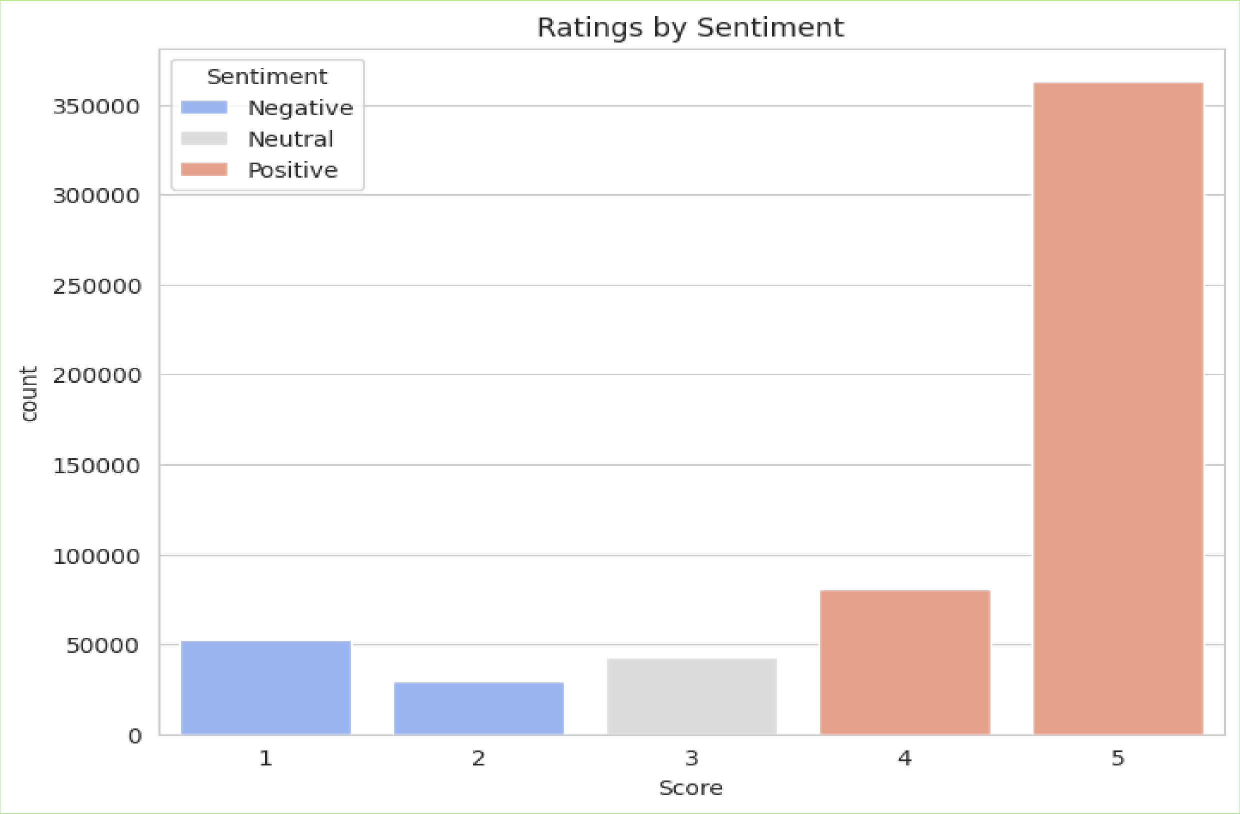


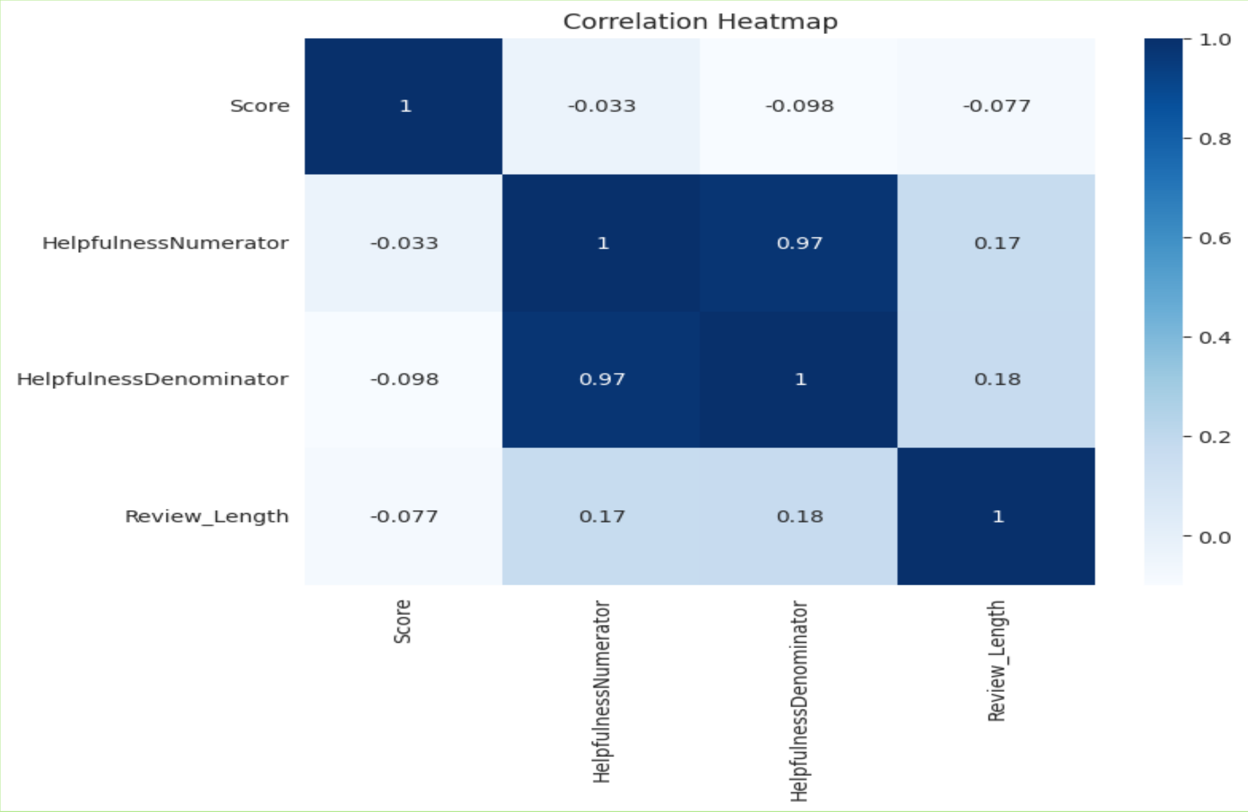

Sentiment Distribution
Number of Reviews
400000
300000
200000
100000
0
Positive
Negative
Neutral
Sentiment

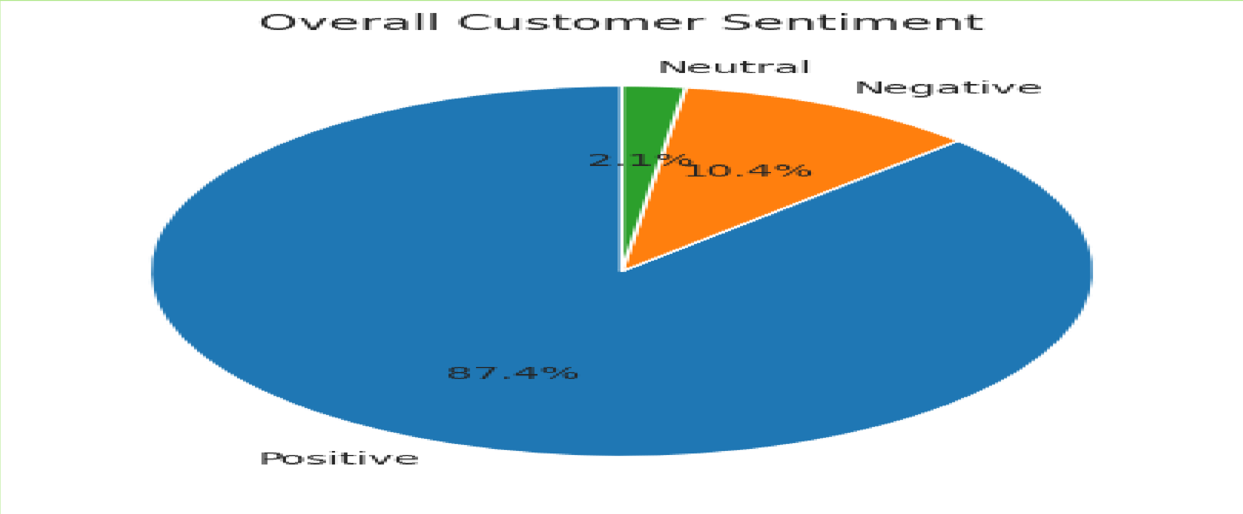

Sentiment Percentage
Positive
78.1%
7.5%
Neutral
14.4%
Negative

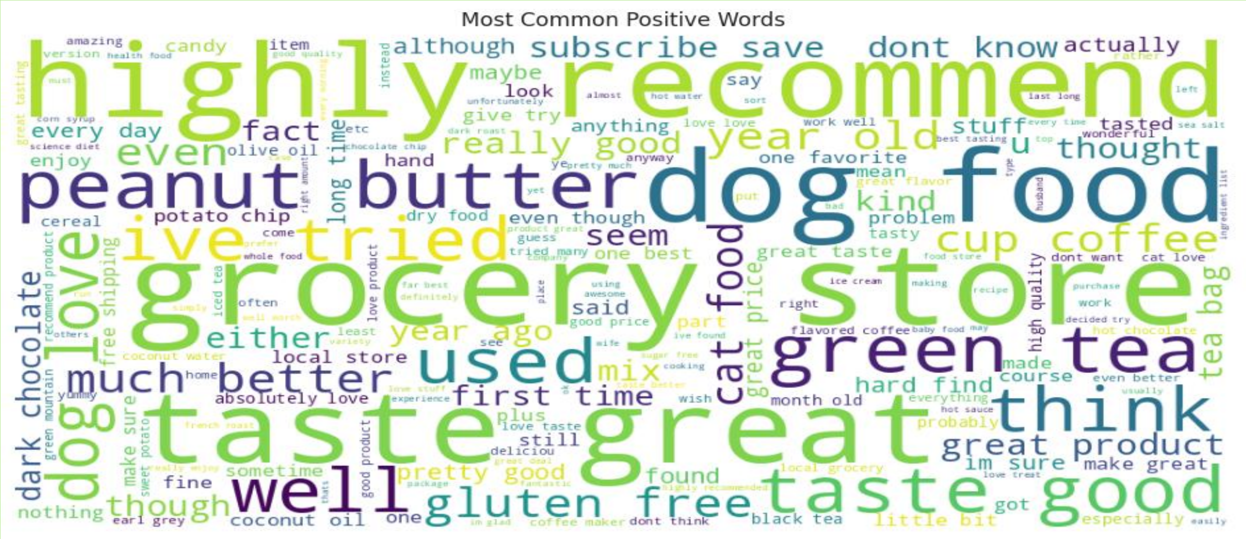

Distribution of Review Length
Frequency
500000
400000
300000
200000
100000
0
0
5000
10000
15000
20000
Characters

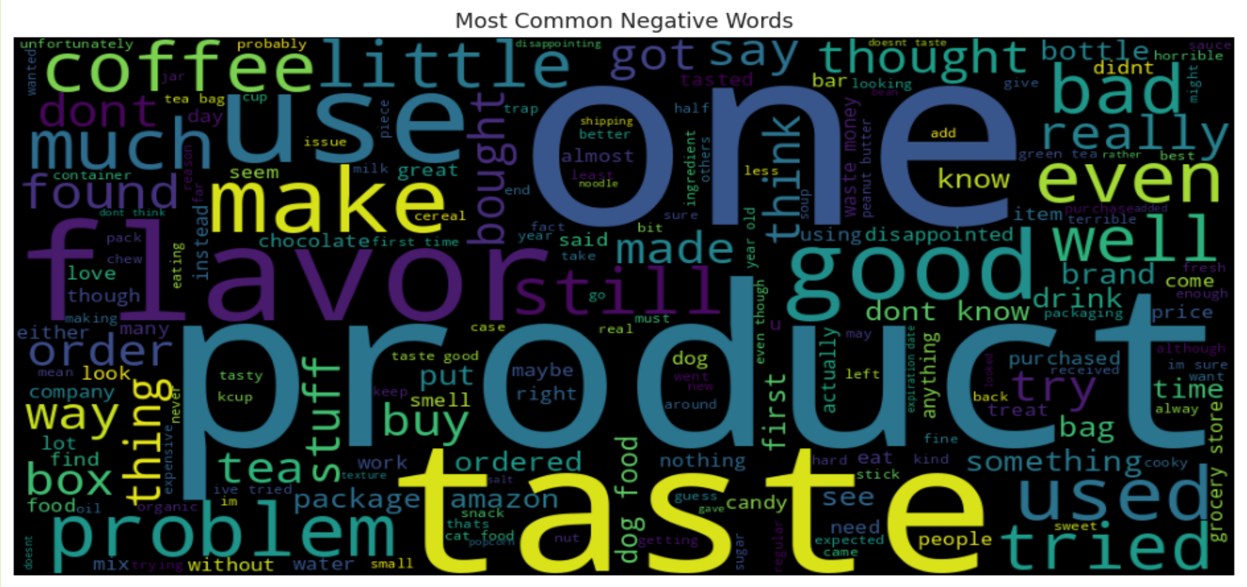

Review Length by Sentiment
Review_Length
20000
15000
10000
5000
0
Positive
Negative
Neutral
Sentiment


Ratings by Sentiment
Sentiment
Negative
Neutral
Positive
350000
300000
250000
200000
150000
100000
50000
0
count
1
2
3
4
5
Score

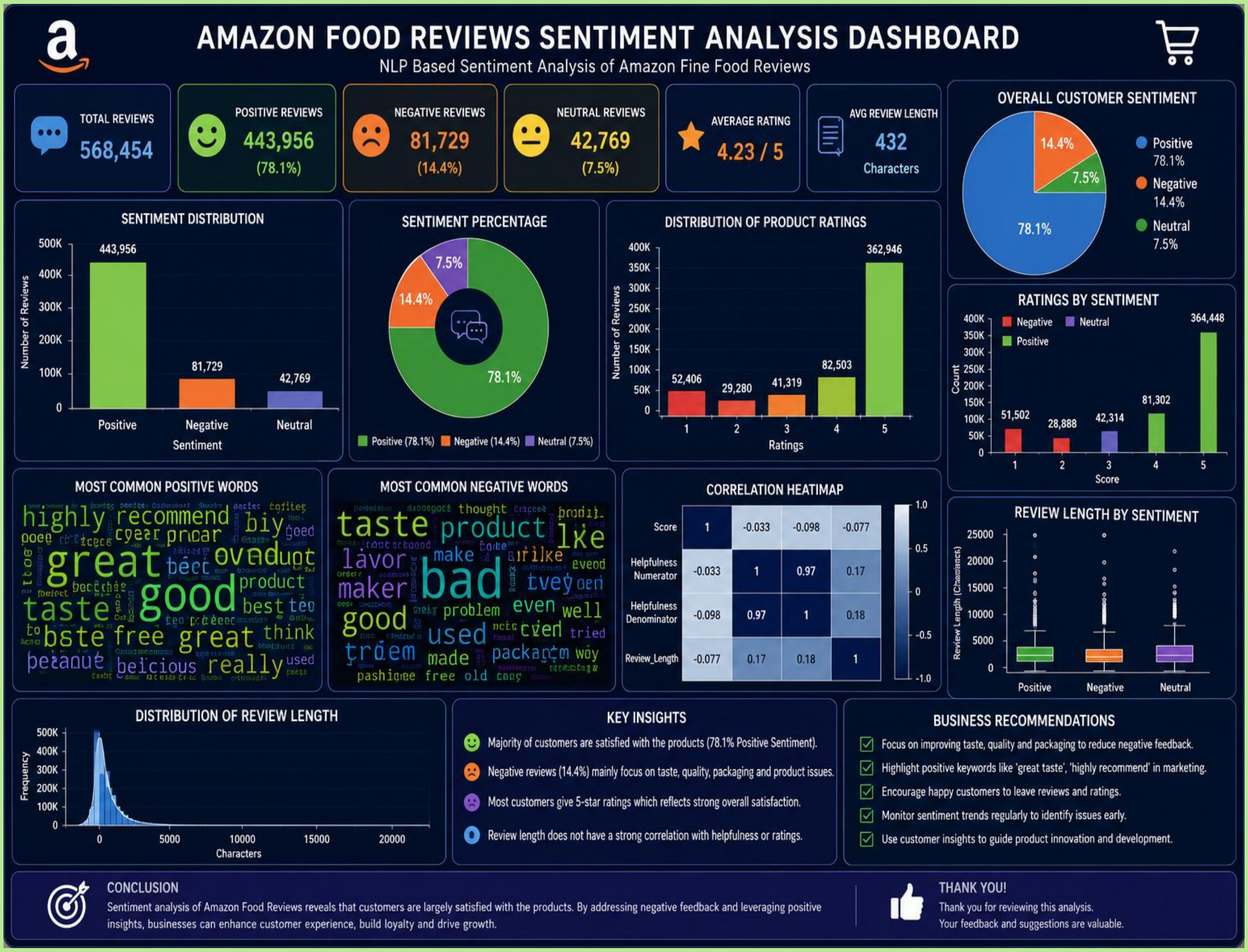

Correlation Heatmap
Score
HelpfulnessNumerator
HelpfulnessDenominator
Review_Length
1
-0.033
-0.098
-0.077
-0.033
1
0.97
0.17
-0.098
0.97
1
0.18
-0.077
0.17
0.18
1
1.0
0.8
0.6
0.4
0.2
0.0
Score
HelpfulnessNumerator
HelpfulnessDenominator
Review_Length


Overall Customer Sentiment
Neutral
Negative
Positive
2.1%
10.4%
87.4%


Most Common Positive Words
highly recommend
grocery store
dog food
peanut butter
taste
great
green tea
taste good
gluten free
think
used
ive tried
love
well
better
much
cup coffee
year old
really good
great product
cat food
dark chocolate
subscribe save
dont know
actually
although
candy
item
amazing


Most Common Negative Words
product
one
taste
flavor
use
coffee
little
make
good
much
found
problem
used
tried
even
well
bad
really
thought
bottle
say
got
dont
still
made
think
bought
box
tea
stuff
thing
way
order
buy
try
time
bag
something
amazon
ordered
package
dog food

AMAZON FOOD REVIEW DASHBOARD



THANK YOU